3D Viewing, Orthographic and Perspective Projection

Transformations in OpenGL

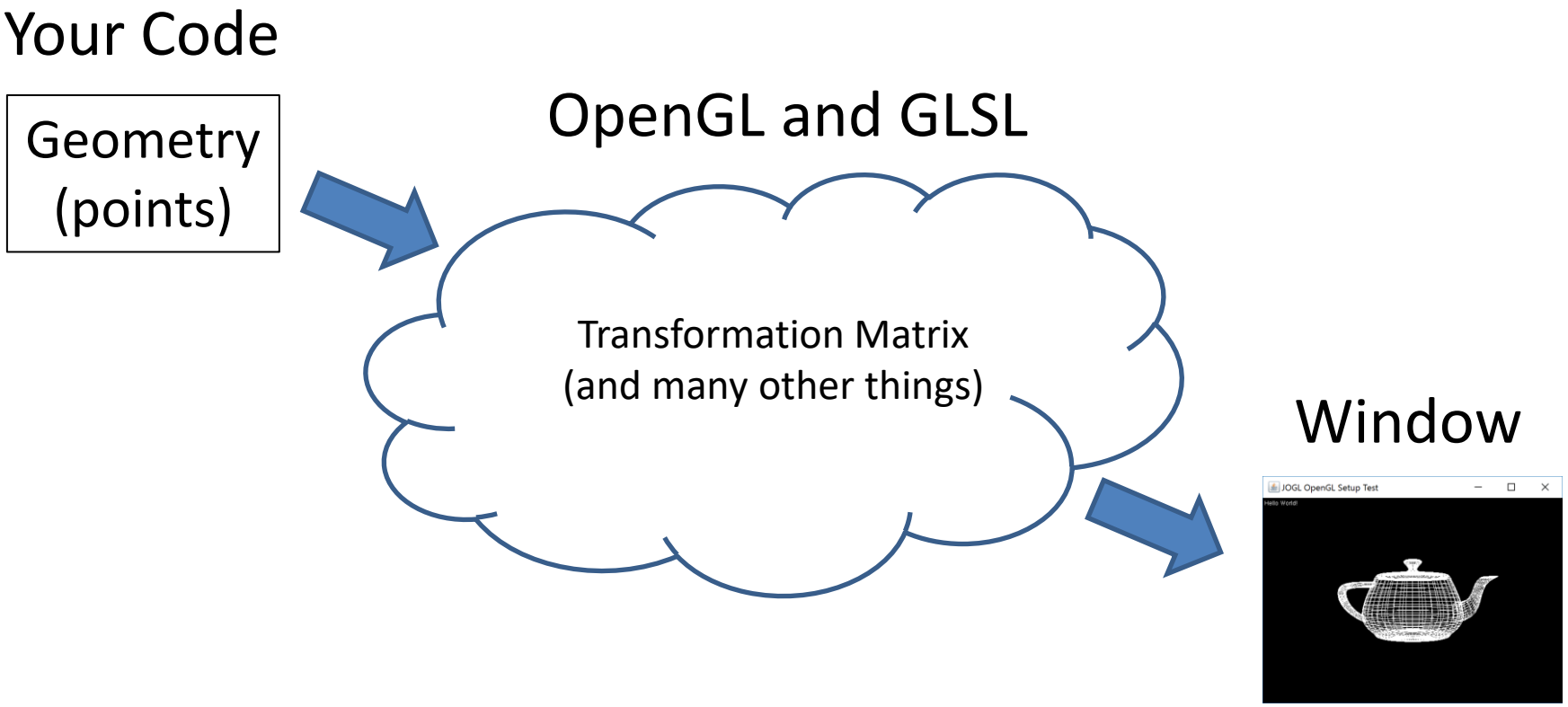
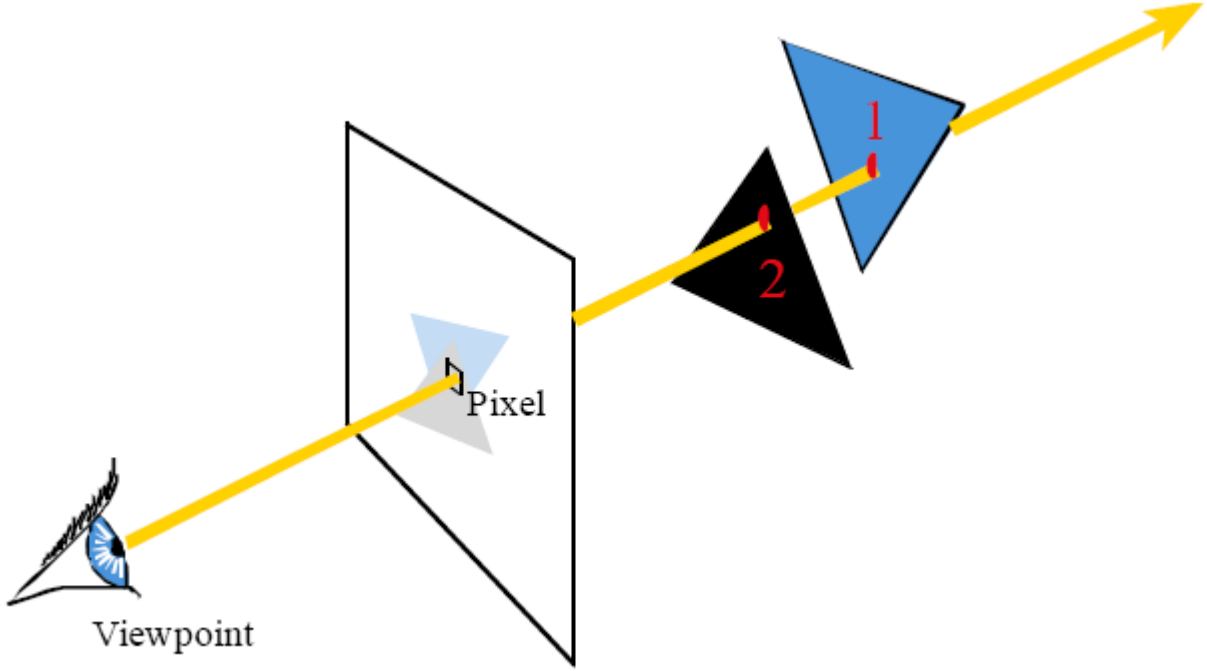


Image order and Object order

- Image order: “backward” approach
  - Start from pixel and ask what part of scene projects to pixel
  - Explicitly construct the ray corresponding to the pixel
- Object order: “forward” approach
  - Start from a point in 3D and compute its projection into the image
- Matrix transformations critical for object order approach (important for both)
  - Combines seamlessly with coordinate transformations used to position camera and model
  - Ultimate goal: single matrix operation to map any 3D point to its correct screen location.

Based on slides by Steve Marschner

Image Order Approach to Viewing

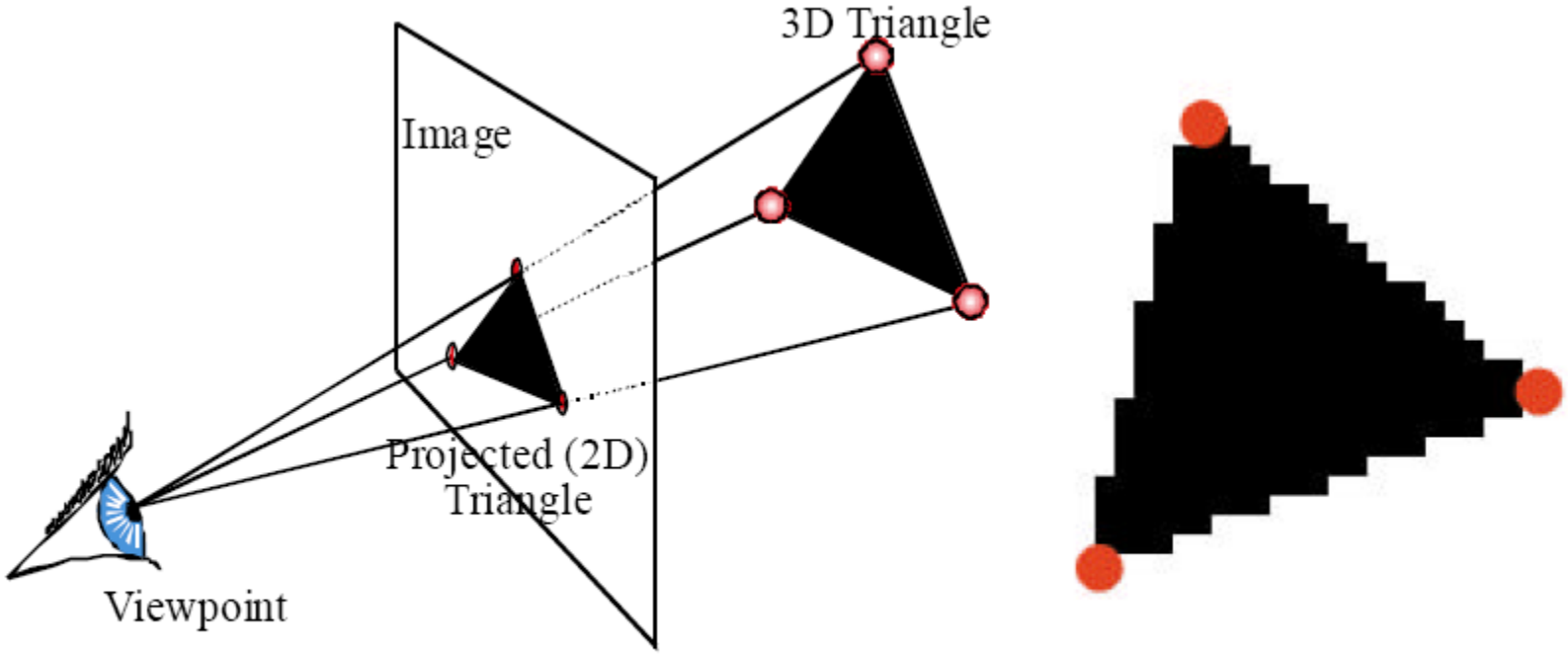


[Durand]

- Ray generation produces rays, not points in scene
- Cast a ray at every pixel and find points of intersection
- This is called ray casting and ray tracing

Based on slides by Steve Marschner

Object Order Approach to Viewing



[Durand]

Projection (left) and rasterization (right) of a triangle.

- Inverting the ray tracing process requires division for the perspective case
- Once triangle vertices are known in screen coordinates the triangle can be filled in (rasterization… more on this later)

Based on slides by Steve Marschner

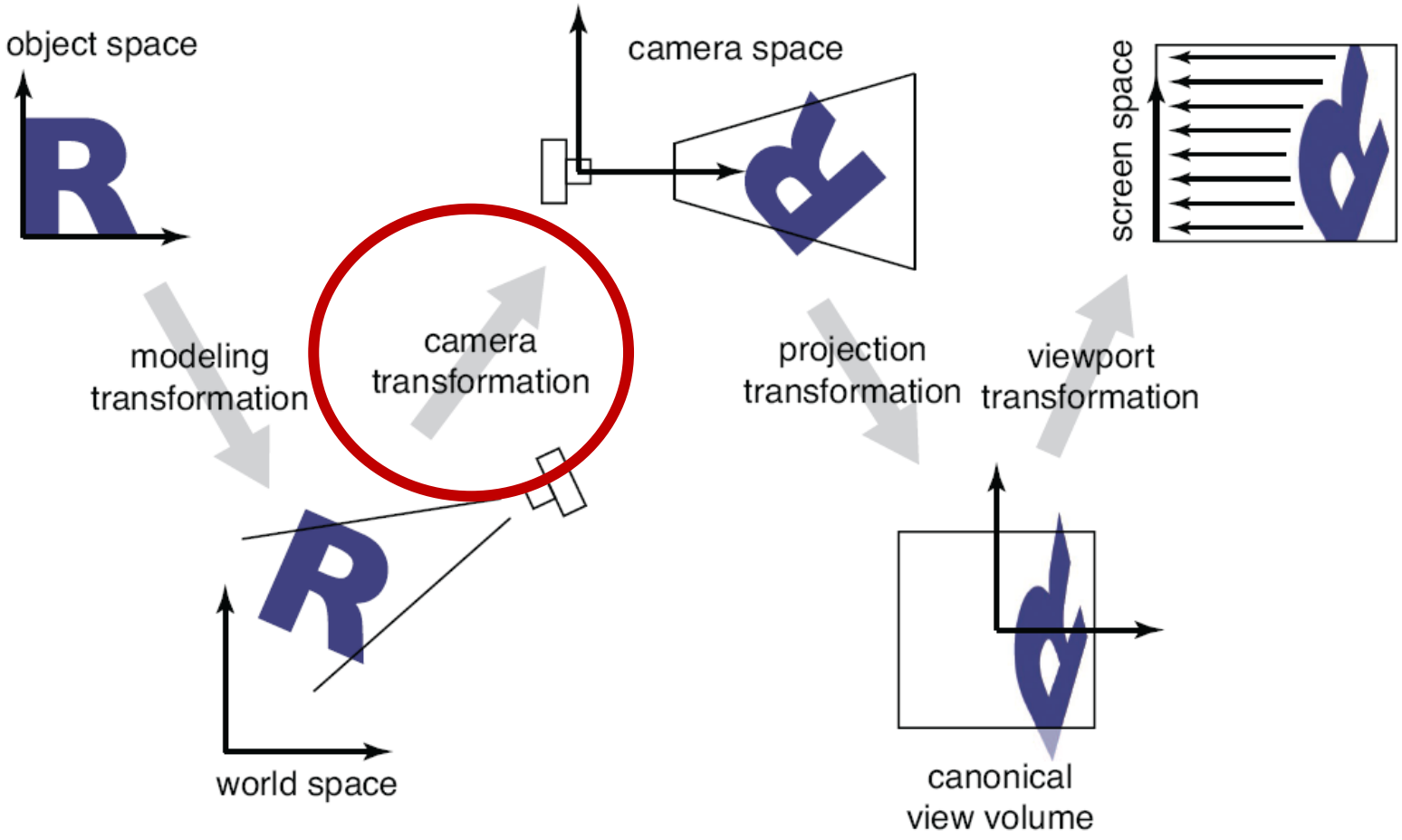
Mathematics of projection

- Always work in eye coordinates
  - Assume eye point at the origin and plane perpendicular to z
- Orthographic case
  - A simple projection: just toss out z
- Perspective case: scale diminishes with z
  - And increases with d

Based on slides by Steve Marschner

Pipeline of transformations

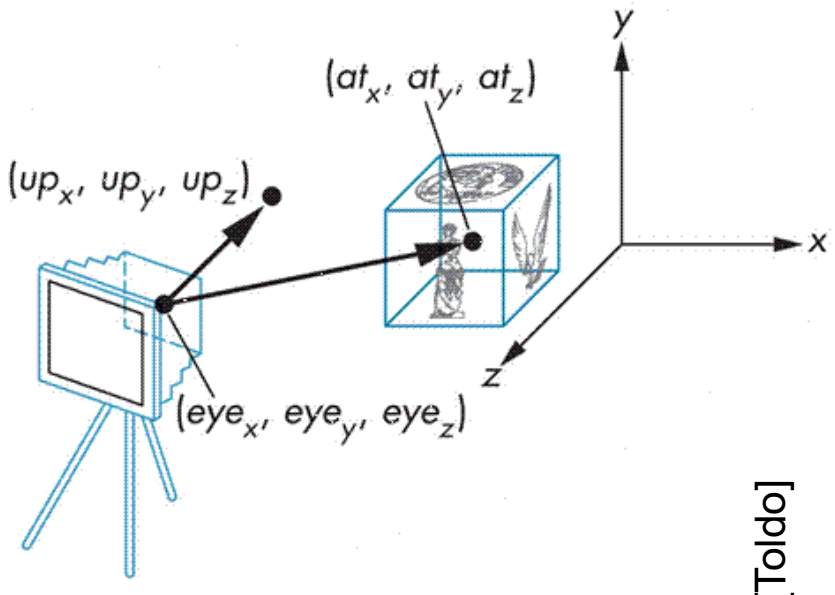
- Standard sequence of transforms



Based on slides by Steve Marschner

Camera / Eye Coordinates

- We have discussed object to world transformations
- Need world to camera transformation
  - In viewing, we typically know:
    - Where the camera is
    - What we want to look at
    - Which way is up
  - Need a rigid transformation (rotation and translation)
  - How many ***degrees of freedom***?
  - How to compute it?



[Toldo]

Based on slides by Steve Marschner

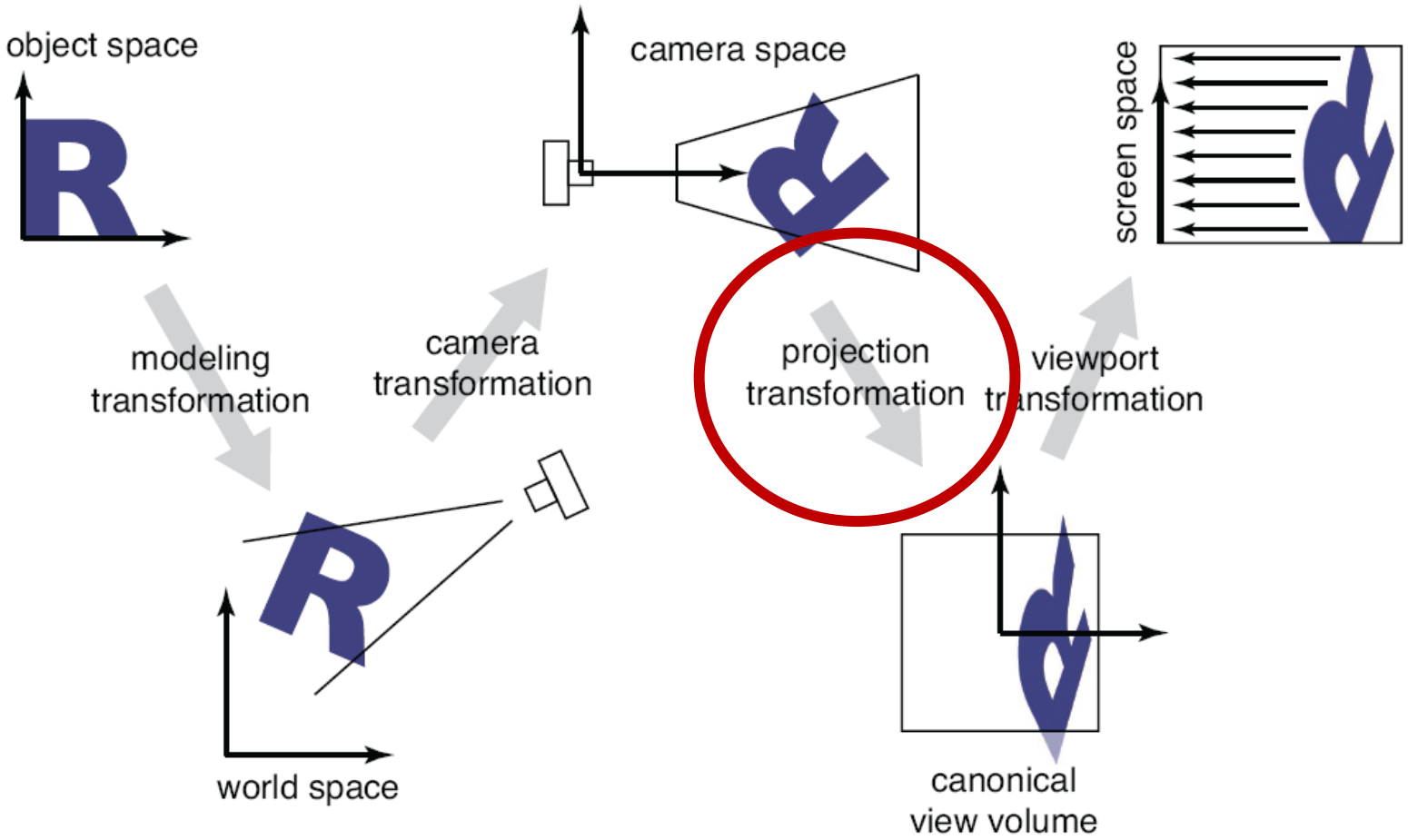
“Lookat” Transformation

- Letters $\boldsymbol{uvn}$ or $\boldsymbol{uvw}$ commonly used for $\boldsymbol{xyz}$ camera axes
- Compute transform from points $\boldsymbol{e}$, $\boldsymbol{l}$, and up vector $\boldsymbol{V}_{up}$ where all quantities are expressed in the world coordinate system
  - $\boldsymbol{e}$ is the eye point, or view reference point
  - $\boldsymbol{l}$ is the look-at point
  - $\boldsymbol{V}_{up}$ is a vector that projects to be vertical in the image plane of the camera
- How can you build a coordinate frame for the camera?
- For the viewing transformation, we want $\begin{pmatrix} \boldsymbol{u} & \boldsymbol{v} & \boldsymbol{w} & \boldsymbol{e} \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1}$

Based on slides by Steve Marschner

Pipeline of transformations

- Standard sequence of transforms

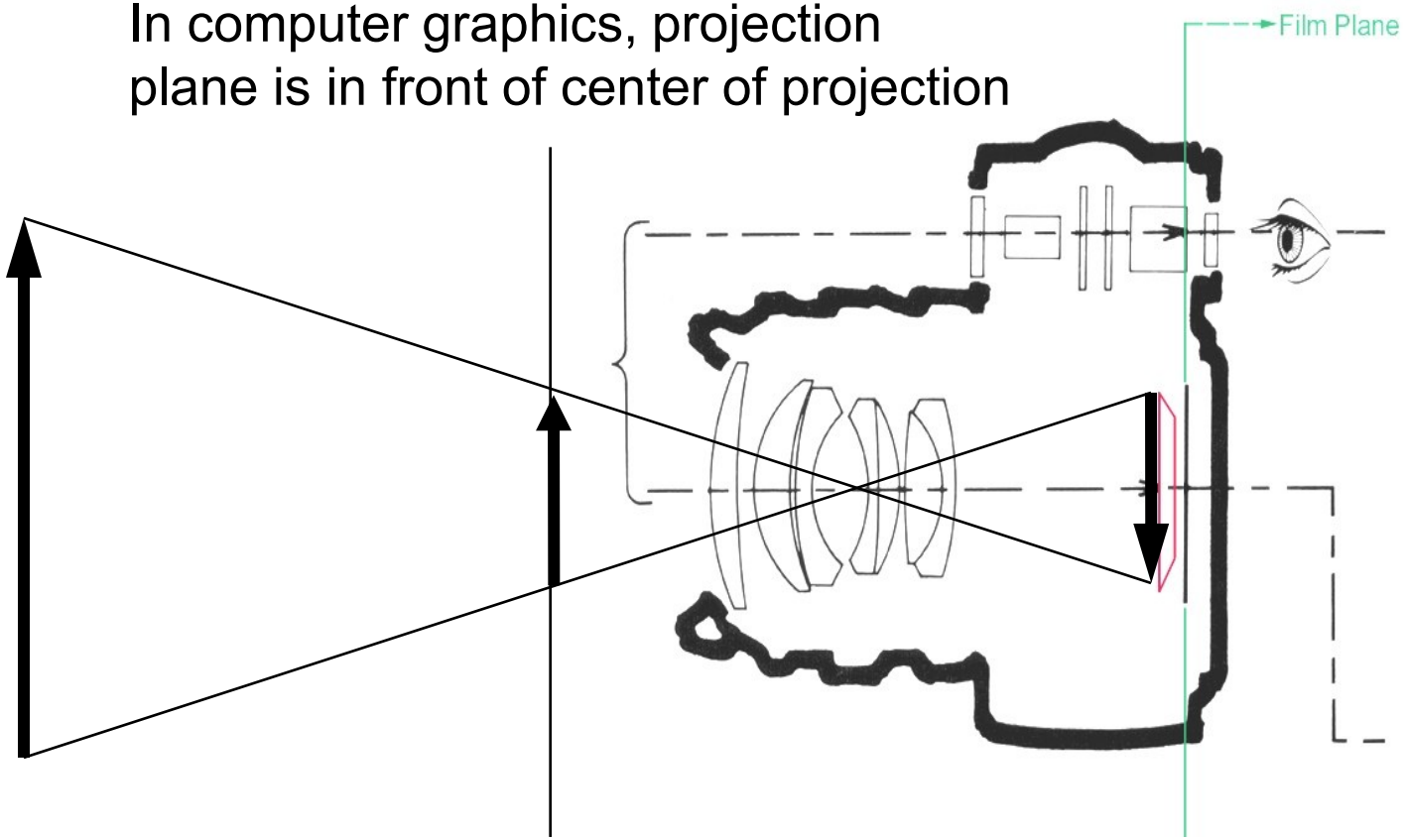


Based on slides by Steve Marschner

Projection

- How to form an image by planar perspective projection?

In computer graphics, projection plane is in front of center of projection



[Source unknown]

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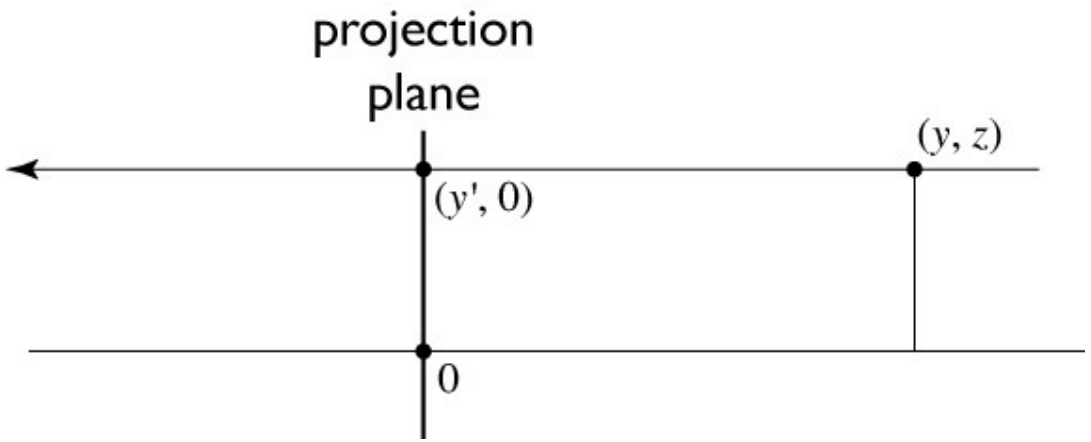
Parallel projection: orthographic



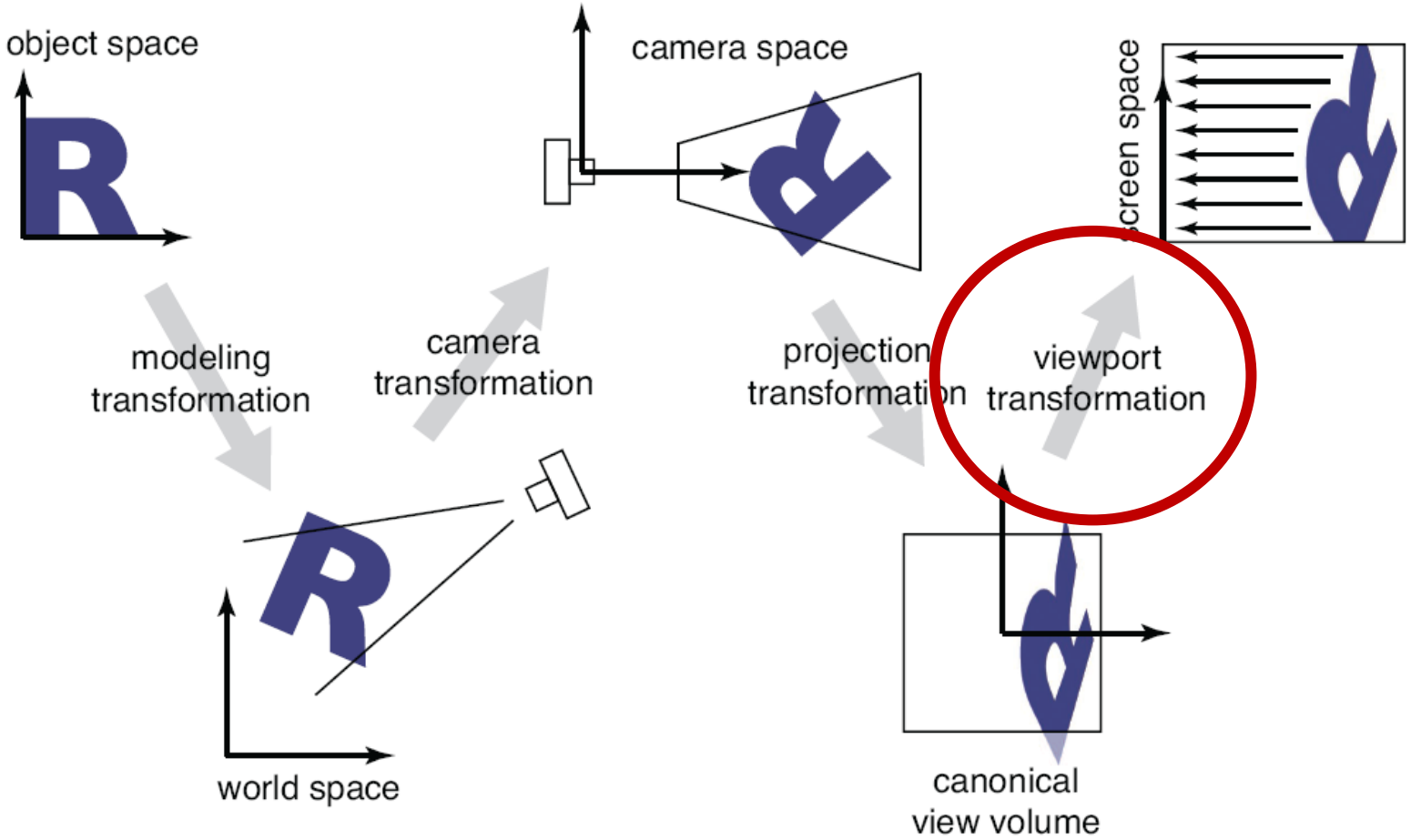
But let us start with something simpler… orthographic projection to implement orthographic, just toss out *z*:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Based on slides by Steve Marschner

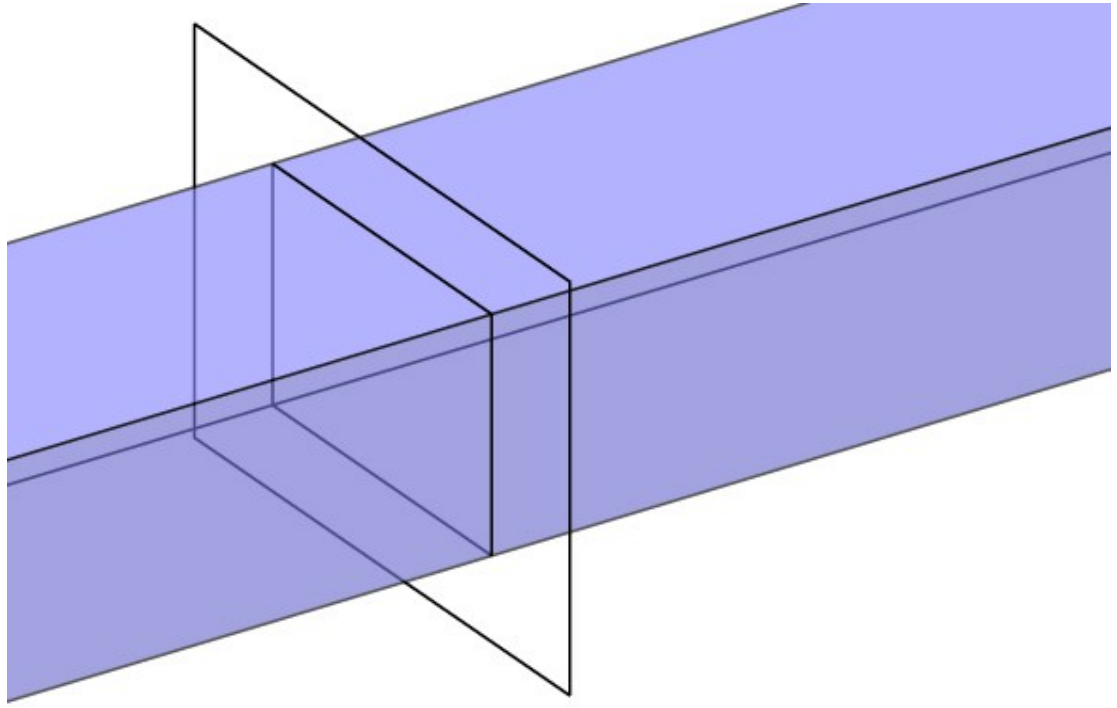
Pipeline of transformations

- Standard sequence of transforms



Based on slides by Steve Marschner

View volume: orthographic



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Viewing a cube of size 2

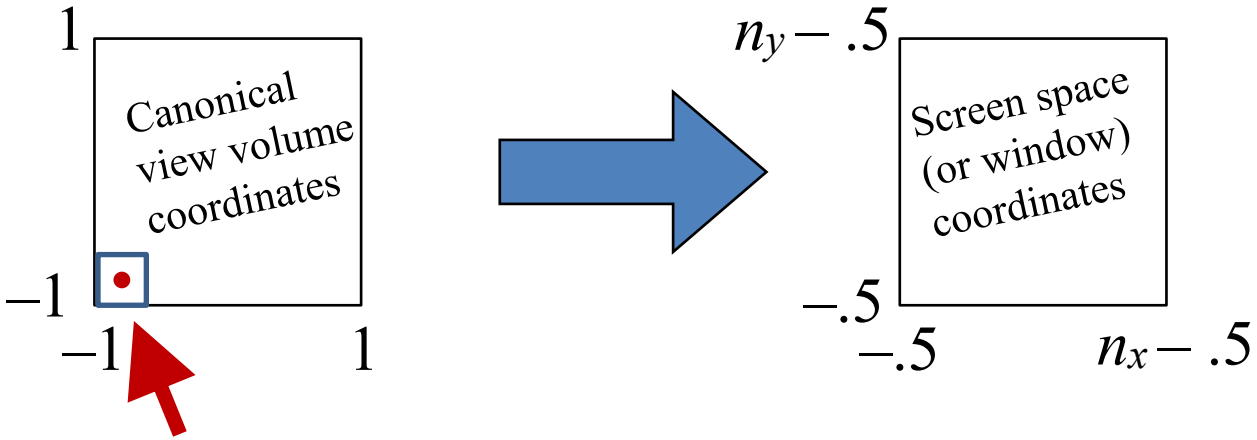
- Start by looking at a restricted case: the canonical view volume
- It is the cube $[-1,1]^3$, viewed from the z direction
- Matrix to project it into a square image in $[-1,1]^2$ is trivial

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Based on slides by Steve Marschner

Viewing a cube of size 2

- To draw in image, need coordinates in pixel units
- Suppose n_x pixels in x direction and n_y pixels in y direction

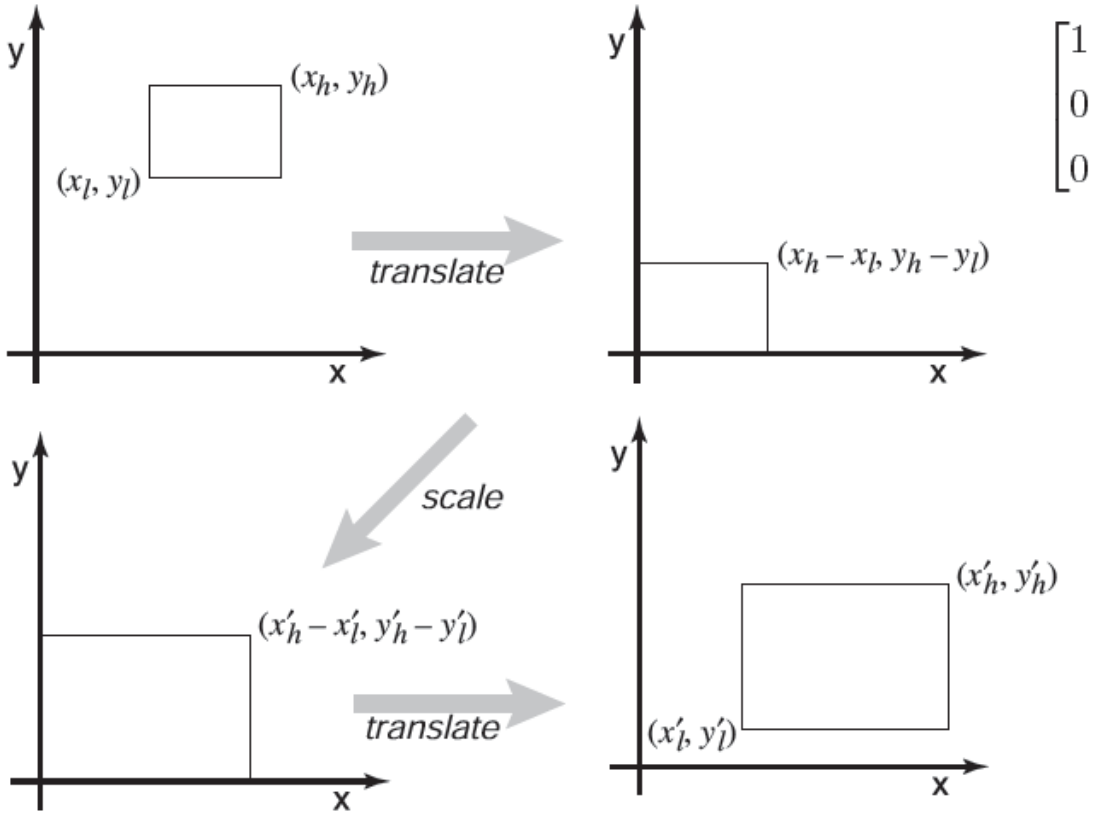


Pixel size in canonical view volume is $2/n_x$ by $2/n_y$
Center is at ½ pixel width and height away from $(-1, -1)$
i.e., $(-1 + 1/n_x, -1 + 1/n_y)$ maps to $(0,0)$ integer pixel location
$(1 - 1/n_x, 1 - 1/n_y)$ maps to $(nx - 1, ny - 1)$ integer pixel location

Based on slides by Steve Marschner

Windowing transforms

- This transformation is worth generalizing



$$\begin{bmatrix} 1 & 0 & x'_l \\ 0 & 1 & y'_l \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{x'_h - x'_l}{x_h - x_l} & 0 & 0 \\ 0 & \frac{y'_h - y'_l}{y_h - y_l} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_l \\ 0 & 1 & -y_l \\ 0 & 0 & 1 \end{bmatrix}$$

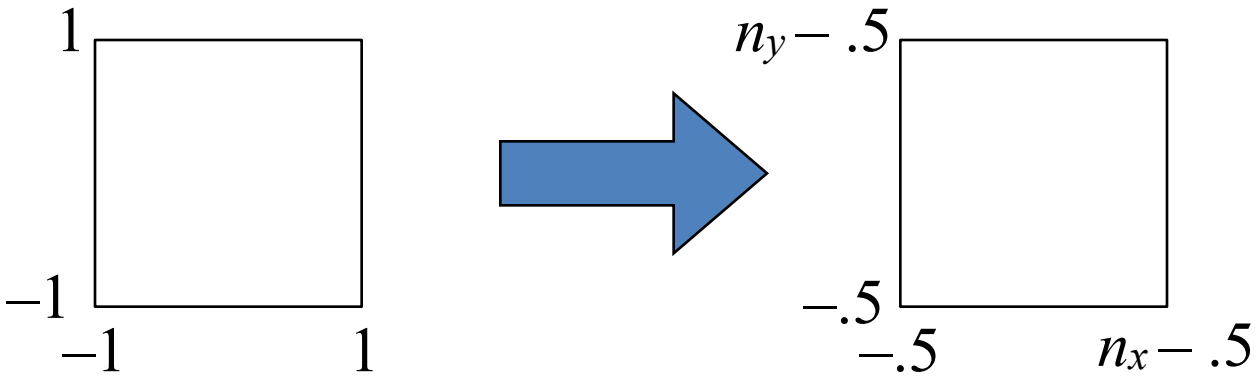
$$\begin{bmatrix} \frac{x'_h - x'_l}{x_h - x_l} & 0 & \frac{x'_l x_h - x'_h x_l}{x_h - x_l} \\ 0 & \frac{y'_h - y'_l}{y_h - y_l} & \frac{y'_l y_h - y'_h y_l}{y_h - y_l} \\ 0 & 0 & 1 \end{bmatrix}$$

Take one axis-aligned rectangle or box to another, a useful transformation chain!

[Shirley3e f. 6-16; eq. 6-6]

Based on slides by Steve Marschner

Viewport transformation



$$\begin{bmatrix} x_{\text{screen}} \\ y_{\text{screen}} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{n_x}{2} & 0 & \frac{n_x - 1}{2} \\ 0 & \frac{n_y}{2} & \frac{n_y - 1}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{canonical}} \\ y_{\text{canonical}} \\ 1 \end{bmatrix}$$

$(-1+1/n_x, -1+1/n_y)$ maps to $(0,0)$
$(1-1/n_x, 1-1/n_y)$ maps to (n_x-1, n_y-1)

In OpenGL 4.0, glViewport sets up this transform

In moderngl python OpenGL bidings, Context.viewport

Should be set on window resize, or in some cases, it is set automatically

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Viewport transformation

- In 3D, carry along *z* for the ride
  - one extra row and column

$$\mathbf{M}_{vp} = \begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x - 1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y - 1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{canonical}} \\ y_{\text{canonical}} \\ 1 \end{bmatrix}$$

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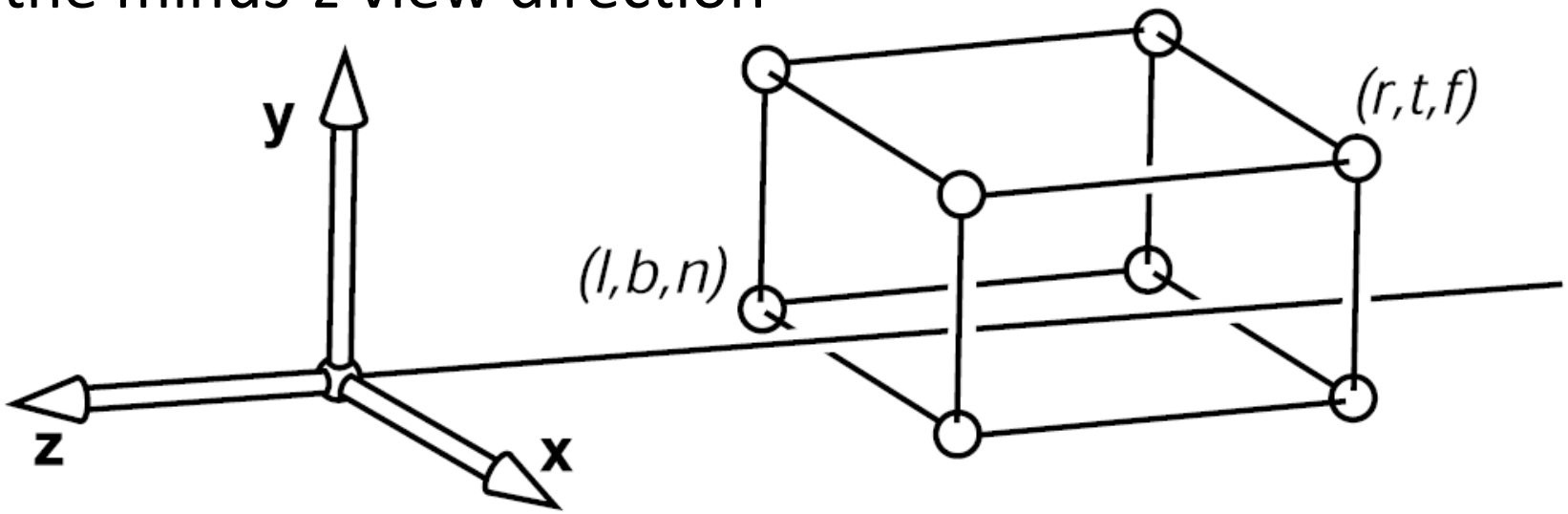
Orthographic projection

- First generalization: different view rectangle
  - retain the minus-z view direction



  - specify view by left, right, top, bottom
  - also near, far

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Clipping planes

- Recall…
  - Object-order rendering considers each object in turn, i.e., forward rendering, rasterization
  - Image-order rendering considers each pixel in turn, i.e., backward rendering, ray tracing
- In object-order systems we **always** use at least two *clipping planes* that further constrain the view volume
  - near plane: parallel to view plane; things between it and the viewpoint will not be rendered
  - far plane: also parallel; things behind it will not be rendered
- These planes are:
  - partly to *remove unnecessary stuff* (e.g., behind the camera)
  - but really to *constrain the range of depths* (we'll see why later)

Based on slides by Steve Marschner

Orthographic projection

- We can implement this by mapping the view volume to the canonical view volume.
- This is just a 3D windowing transformation!

$$\begin{bmatrix} \frac{x'_h - x'_l}{x_h - x_l} & 0 & 0 & \frac{x'_l x_h - x'_h x_l}{x_h - x_l} \\ 0 & \frac{y'_h - y'_l}{y_h - y_l} & 0 & \frac{y'_l y_h - y'_h y_l}{y_h - y_l} \\ 0 & 0 & \frac{z'_h - z'_l}{z_h - z_l} & \frac{z'_l z_h - z'_h z_l}{z_h - z_l} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{M}_{orth} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

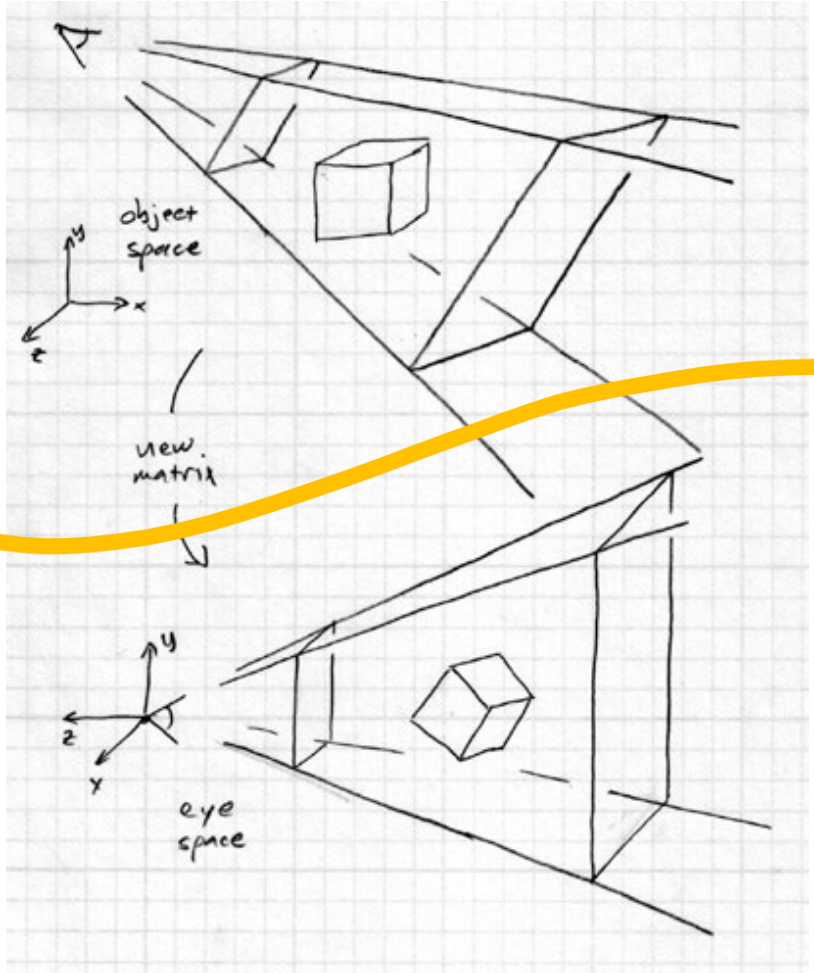
Based on slides by Steve Marschner

Camera and modeling matrices

- We worked out all the preceding transforms starting from eye coordinates
  - before we do any of this, we must transform into that space
- Transform from world (canonical) to eye space is called the viewing matrix
  - Easy for us to compute the matrix F_c which takes us from eye space to canonical space, but here we use the inverse F_c^{-1}
- Remember that geometry would originally have been in the object's local coordinates;
  - Transform into world coordinates is called the modeling matrix, M_m
- Note some systems (e.g., OpenGL) combine the two into a modelview matrix and just skip world coordinates

Based on slides by Steve Marschner

Viewing transformation



the camera matrix rewrites all coordinates in eye space

Based on slides by Steve Marschner

Orthographic transformation chain

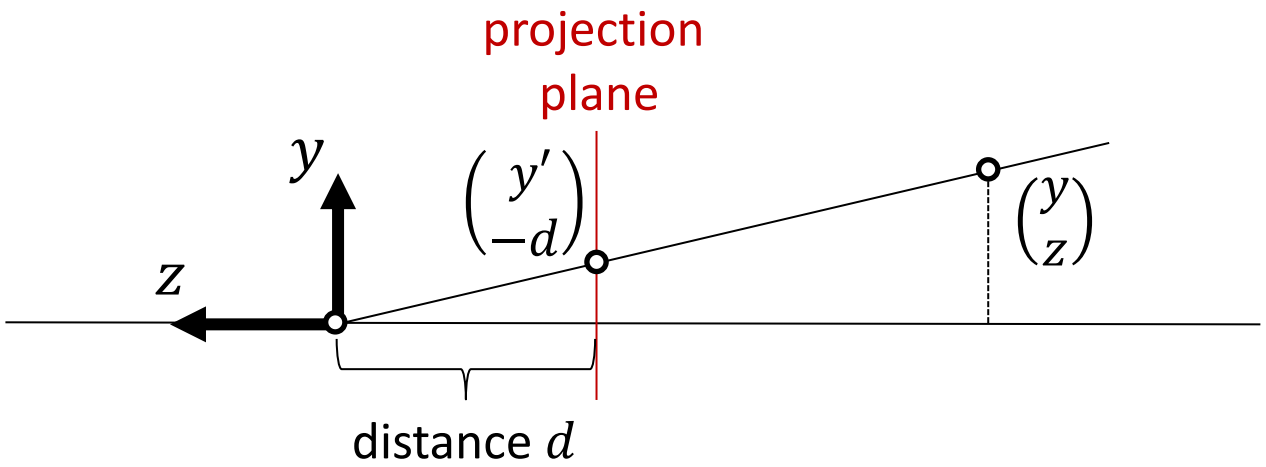
- Start with coordinates in object's local coordinates, p_o
  - Transform into world coordinates (modeling transform, M_m)
  - Transform into eye coordinates (camera transform, $M_{cam} = F_c^{-1}$)
  - Orthographic projection, M_{orth}
  - Viewport transform, M_{vp}

$$p_s = M_{vp}\, M_{orth}\, M_{cam}\, M_m\, p_o$$

$$\begin{bmatrix} x_s \\ y_s \\ z_c \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{OpenGL set with glViewport}} \underbrace{\begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & -\frac{n+f}{n-f} \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Projection Matrix}} \underbrace{\begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} & \mathbf{e} \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1}}_{\text{Viewing Matrix}} \underbrace{\mathbf{M}_{\mathrm{m}}}_{\text{Modeling Matrix}} \begin{bmatrix} x_o \\ y_o \\ z_o \\ 1 \end{bmatrix}$$

Based on slides by Steve Marschner

Planar Perspective projection



With the Projection plane at distance d, note the similar triangles:

$$\frac{y'}{d} = \frac{y}{-z}$$
$$y' = -dy/z$$

- y' is the foreshortened version of y, that is, it is smaller by a factor $-d/z$
- We can think of dividing by negative z so that y' has the same sign as y!

Based on slides by Steve Marschner

Homogeneous coordinates revisited

- Perspective requires division
  - that is not part of affine transformations
  - in affine, parallel lines stay parallel
    - therefore no vanishing point
    - therefore no rays converging on viewpoint
- “True” purpose of homogeneous coords: projection

Based on slides by Steve Marschner

Homogeneous coordinates revisited

- Introduced w = 1 coordinate as a placeholder

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

  – used as a convenience for unifying translation with linear

- Can also allow arbitrary non-zero w

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \sim \begin{bmatrix} wx \\ wy \\ wz \\ w \end{bmatrix}$$

Based on slides by Steve Marschner

Implications of w

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \sim \begin{bmatrix} wx \\ wy \\ wz \\ w \end{bmatrix}$$

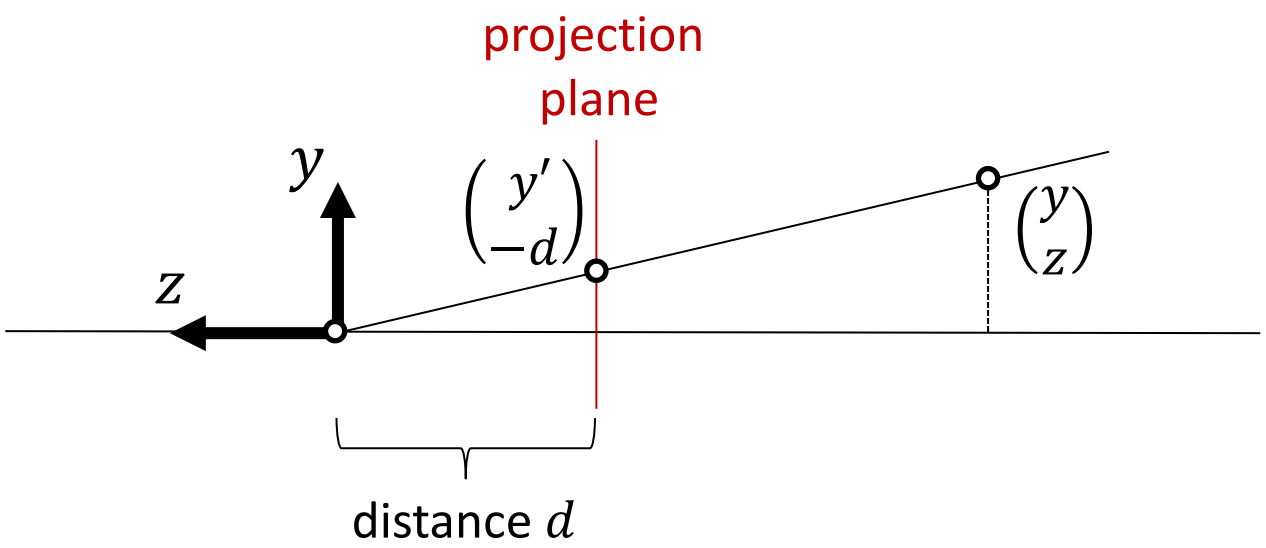
- All scalar multiples of a 4-vector are equivalent
- When w is not zero, can divide by w
  - Therefore, these points represent “normal” affine points
- When w is zero, it’s a point at infinity, i.e., a direction
  - Can think of this as the point where parallel lines intersect
  - Can also think of it as the vanishing point

Based on slides by Steve Marschner

Perspective Projection

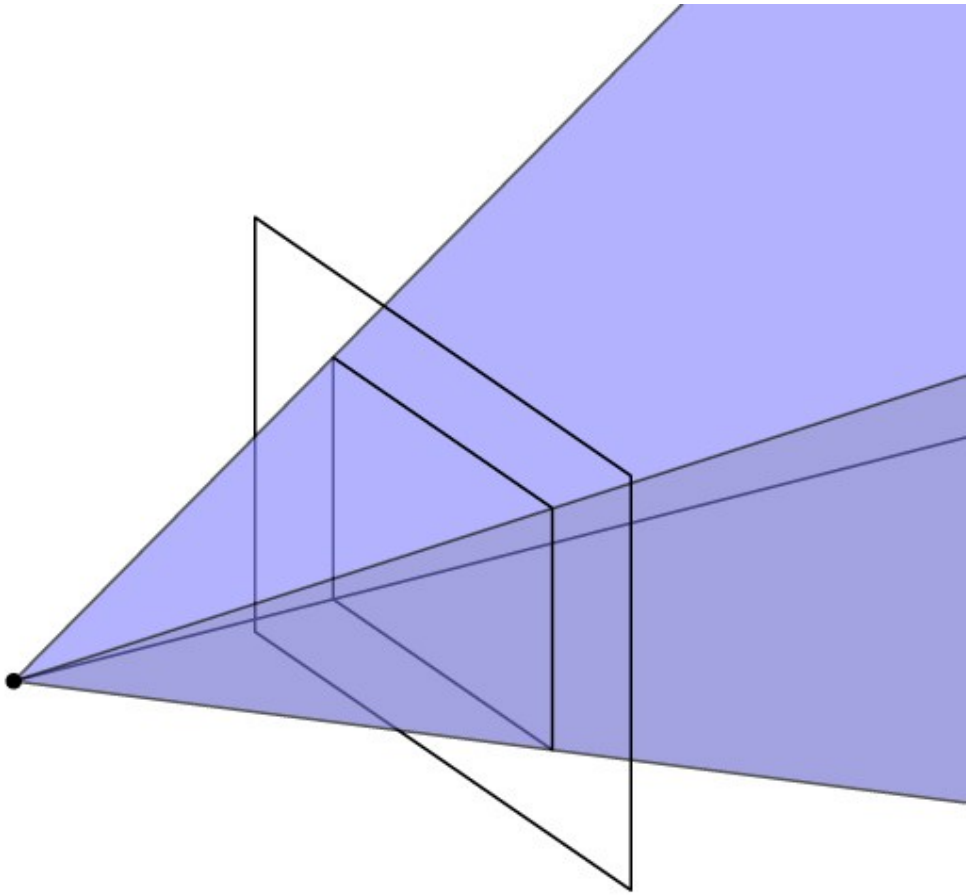
- To implement perspective, put $-z$ into w

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} -dx/z \\ -dy/z \\ 1 \end{pmatrix} \sim \begin{pmatrix} dx \\ dy \\ -z \end{pmatrix} = \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$



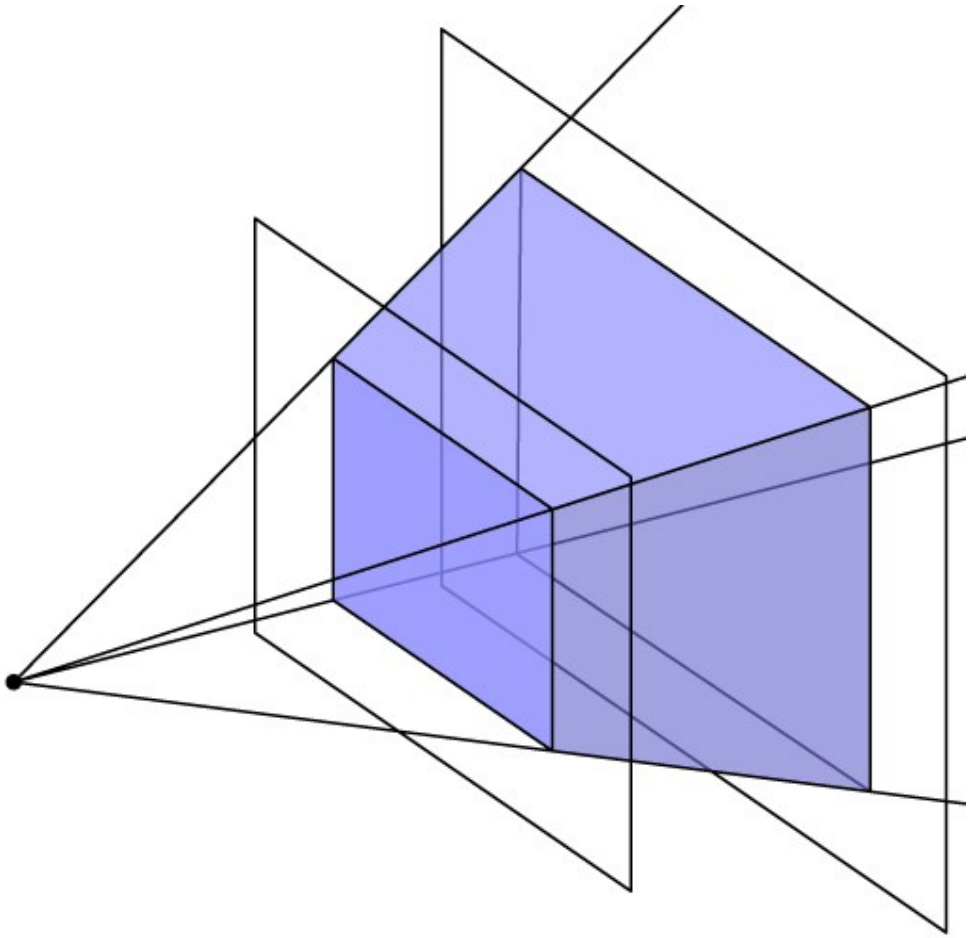
Based on slides by Steve Marschner

View volume: perspective



Based on slides by Steve Marschner

View volume: perspective (clipped)



Based on slides by Steve Marschner

Perspective Projection

Carrying depth through perspective

- Perspective has a varying denominator,
  - We can't preserve depth!
- As a compromise we can preserve depth order, and depth on near and far planes
  - Let n and f be the ***distance*** (+ve) to the near and far plane
  - Let the projection plane be at distance n (That is $d = n$)

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} \sim \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \\ -z \end{pmatrix} = \begin{pmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

  - Choose a and b so that $z'(-n) = -n$ and $z'(-f) = -f$.

Based on slides by Steve Marschner

Carrying depth through perspective

$$z' = \frac{az + b}{-z}$$

$$(-n) = \frac{a(-n) + b}{-(-n)}$$ Plug in $-n$ for z

$$-n^2 = b - an$$ Let this be Equation 1

$$-f^2 = b - af$$ Have similar expression for f

$$-(n^2 - f^2) = -a(n - f)$$ Subtract previous two lines

$$(n + f) = a$$

$$-n^2 = b - (n + f)n$$ Replace a in Equation 1

$$-n^2 = b - n^2 - nf$$

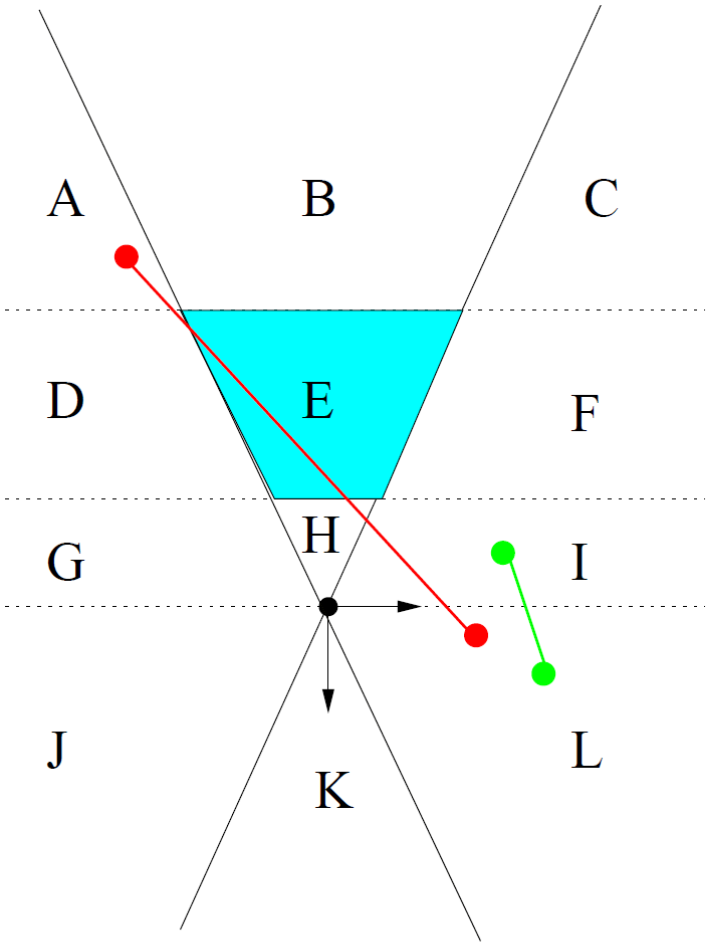
$$nf = b$$

Based on slides by Steve Marschner

Questions

- Questions to better understand the action of a 4x4 homogenous planar perspective transformation matrix
  - What happens to vectors $(x, y, z, 0)$?
    - They map to plane z=near+far
  - What happens to the origin $(0,0,0, w)$?
    - Maps to $(0,0, nfw, 0)$
  - What happens to points on plane **with z normal** at center of projection ?
    - Map to points at infinity (all vectors come from the homogeneous representations of points on this plane at center of projection)
  - What happens to lines through the origin ?
    - They become parallel
  - What happens to points at $z = (-n + -f)/2$?
    - $(n^2 + f^2)/(n + f)$, that is, z is not preserved !! However, order is! (has implications for z depth precision)
  - What happens to points behind the camera ?
    - It goes in front of the camera

Aside: What goes where?



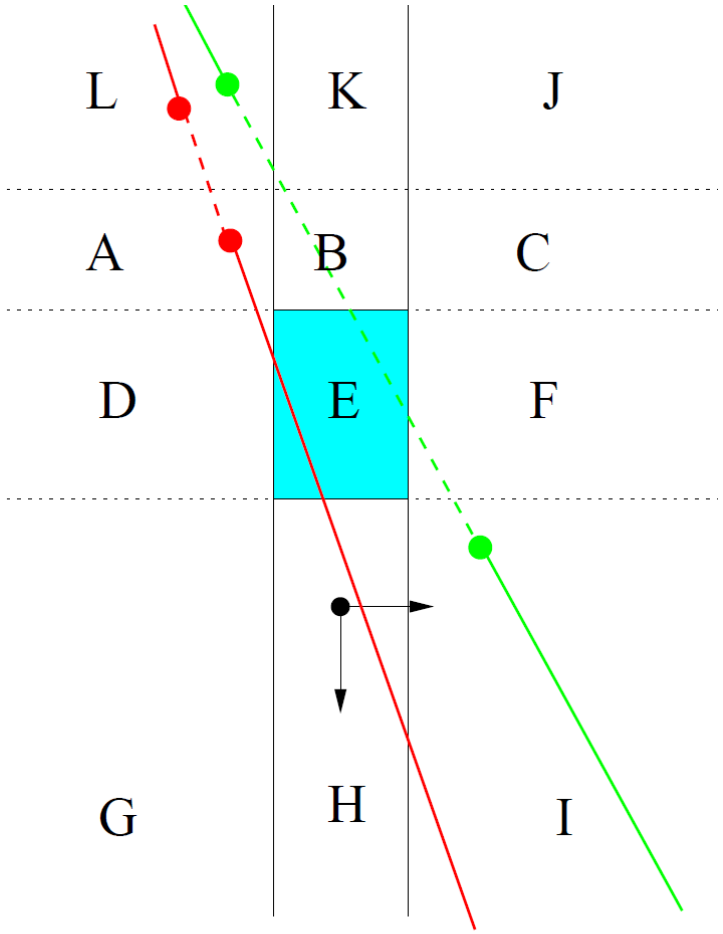
camera coordinates

$$z = -n - f$$

$$z = -f$$

$$z = -n$$

$$z = 0$$



(non−normalized) projection coordinate

This has implications for clipping (i.e., discarding) geometry!

Based on slides by Steve Marschner

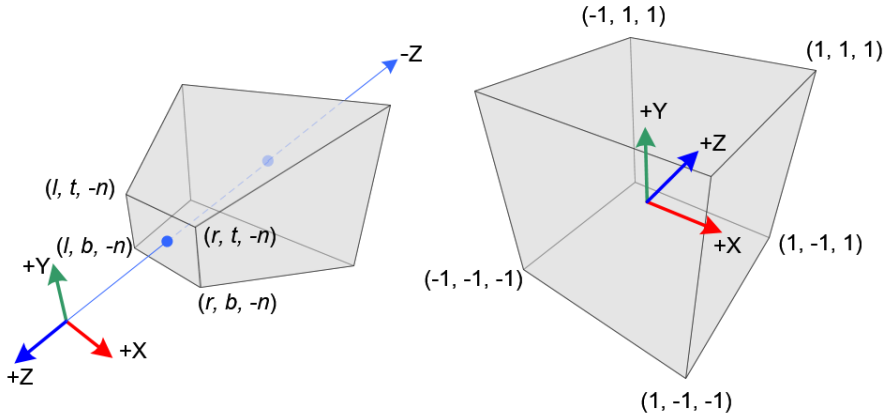
$[z_l, z_h]$ to $[z'_l, z'_h]$, i.e., $[-n, -f]$ to $[-1,1]$

- Another windowing transform
- Convention has points on near plane map to -1 and points on far plane map to 1
- Note that this swaps handedness

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \dfrac{z'_h - z'_l}{z_h - z_l} & \dfrac{z'_l z_h - z'_h z_l}{z_h - z_l} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \dfrac{1-(-1)}{(-f)-(-n)} & \dfrac{(-1)(-f)-(1)(-n)}{(-f)-(-n)} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \dfrac{2}{n-f} & \dfrac{f+n}{n-f} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



Based on slides by Steve Marschner

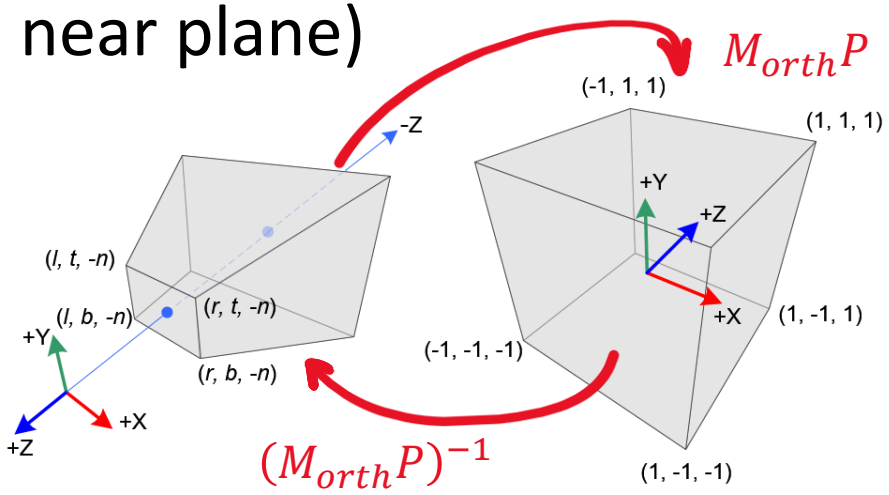
Window transform for x and y too

- Perspective projection must produce points in the canonical view volume, so combine with orthographic projection matrix (windowing transform)
- Use left right top and bottom (on near plane)

$$M_{per} = M_{orth}P$$

$$= \begin{pmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & \frac{f+n}{n-f} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & n+f & nf \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2n}{r-l} & 0 & \frac{r+l}{r-l} & 0 \\ 0 & \frac{2n}{t-b} & \frac{t+b}{t-b} & 0 \\ 0 & 0 & \frac{n+f}{n-f} & \frac{2nf}{n-f} \\ 0 & 0 & -1 & 0 \end{pmatrix}$$



Based on slides by Steve Marschner

Perspective transformation chain

- Transform into world coordinates (modeling transform, M_m)
- Transform into eye coords (camera transform, $M_{cam} = F_c^{-1}$)
- Perspective matrix, P
- Orthographic projection, M_{orth}
- Viewport transform, M_{vp}

$$p_s = M_{vp}\, M_{orth}\, P\, M_{cam}\, M_m\, p_o$$

$$\begin{bmatrix} x_s \\ y_s \\ z_c \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} \frac{n_x}{2} & 0 & 0 & \frac{n_x-1}{2} \\ 0 & \frac{n_y}{2} & 0 & \frac{n_y-1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}} \boxed{\begin{pmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{n-f} & \frac{f+n}{n-f} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & n+f & nf \\ 0 & 0 & -1 & 0 \end{pmatrix}} \boxed{\mathbf{M}_{\mathrm{cam}}\mathbf{M}_{\mathrm{m}}} \begin{bmatrix} x_o \\ y_o \\ z_o \\ 1 \end{bmatrix}$$

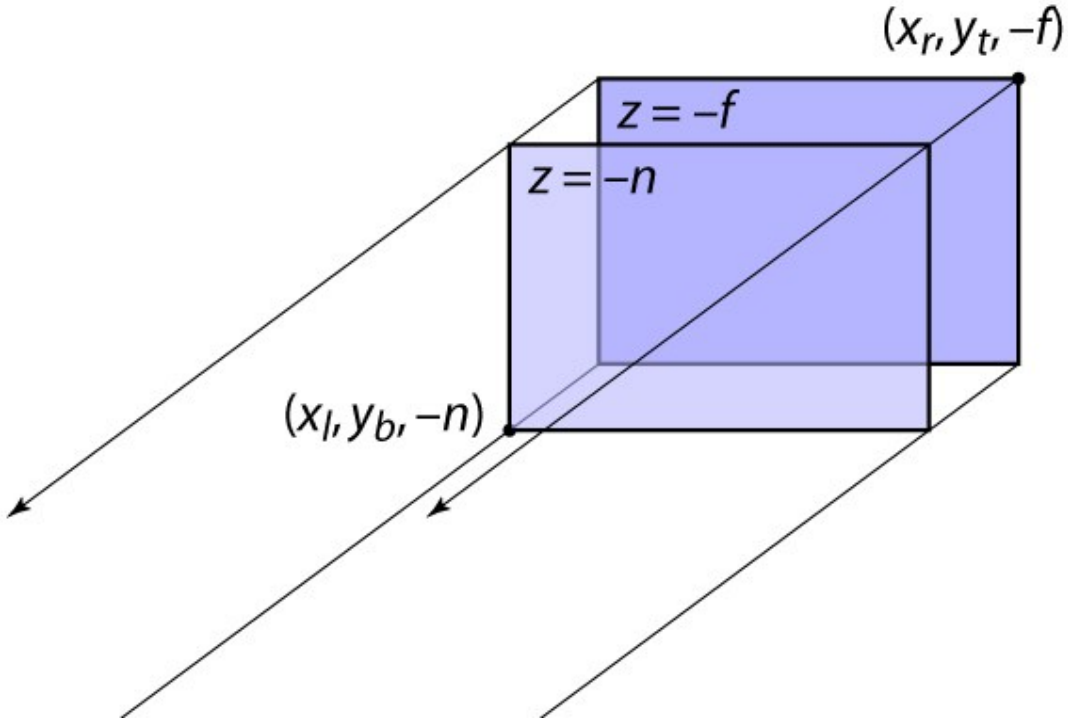
OpenGL set with glViewport

Projection Matrix

Viewing Matrix and Modeling Matrix

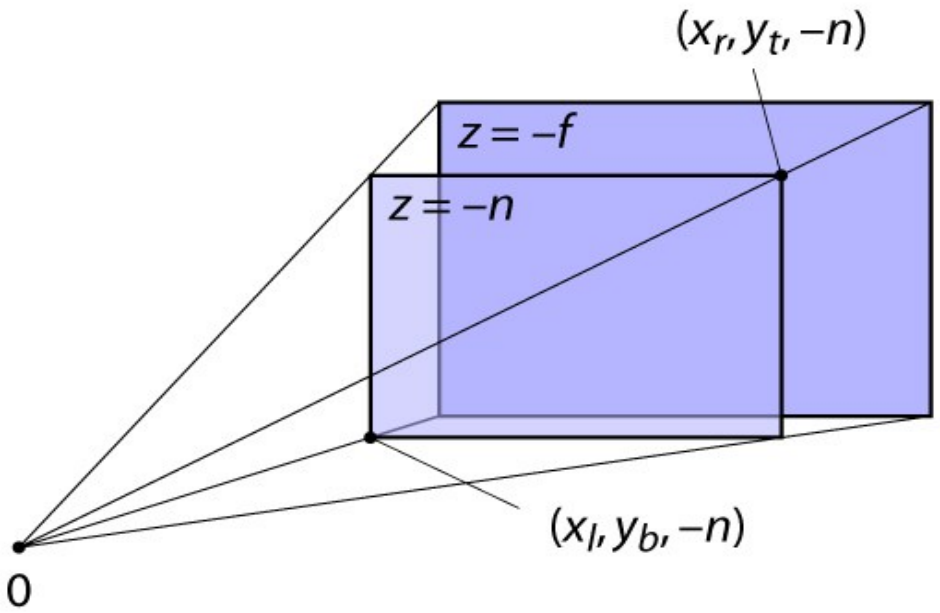
Based on slides by Steve Marschner

OpenGL view frustum: orthographic



Based on slides by Steve Marschner

OpenGL view frustum: perspective



- We can describe a frustum based on
  - field of view in the y direction and the aspect ratio, or
  - the left right bottom and top of the near plane rectangle
- Recall we use distance to near and far, thus the negative z coordinates in the image above.

Based on slides by Steve Marschner

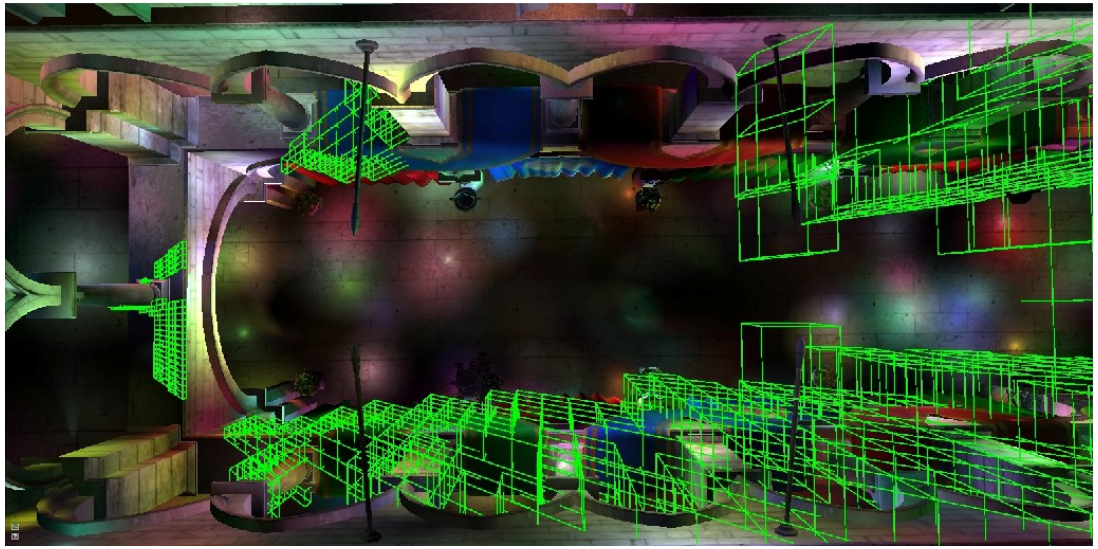
Frustum applications

- Shifted perspective

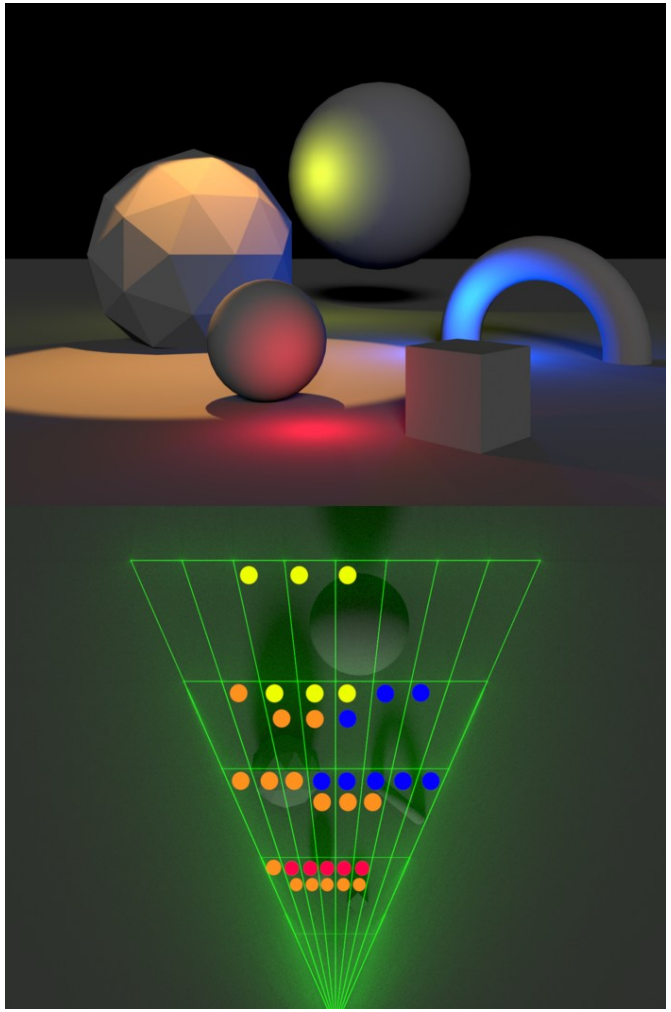
Based on slides by Steve Marschner

Frustum applications

- Tiled rendering
  - Clustered deferred and forward shading [Olsen et al. HPG 2012]
  - Less common: render very high resolutions



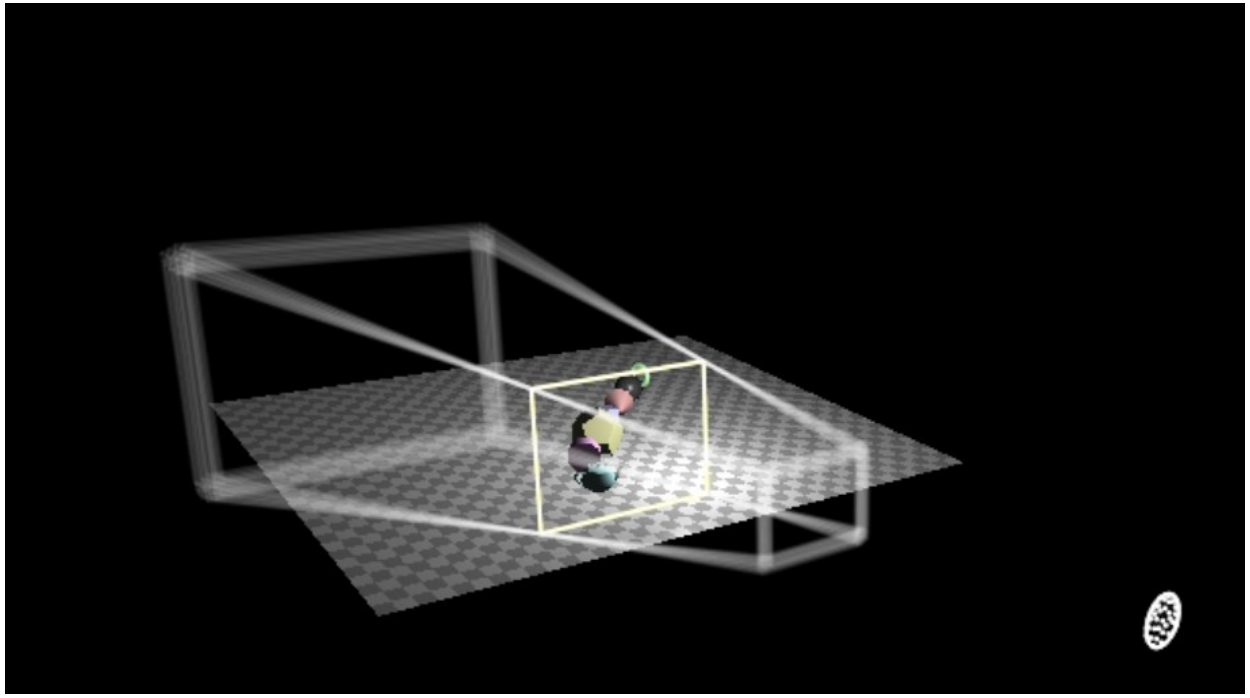
Ortegren 2015 Clustered Shading : Assigning arbitrarily shaped convex light volumes using conservative rasterization

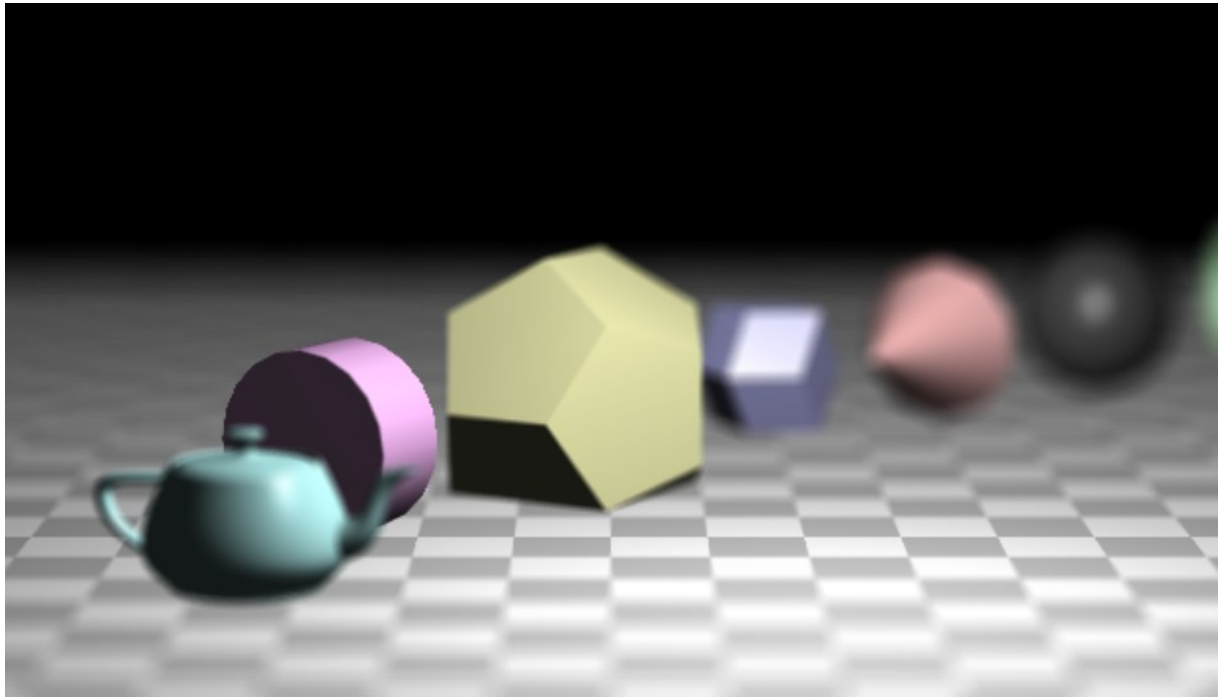


http://www.aortiz.me/2018/12/21/CG.html#tiled-shading--forward

Frustum applications

- Depth of field
  - Accumulating multiple render passes (not a modern approach)





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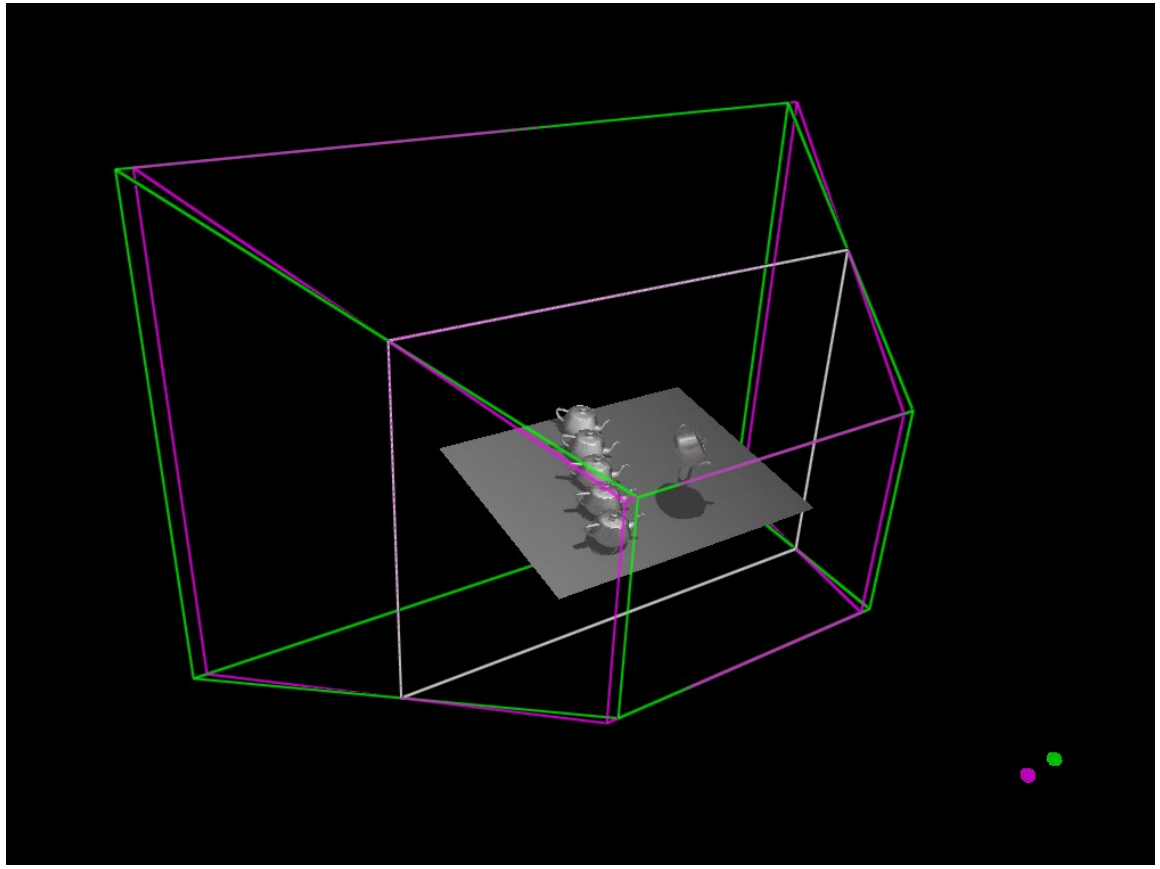
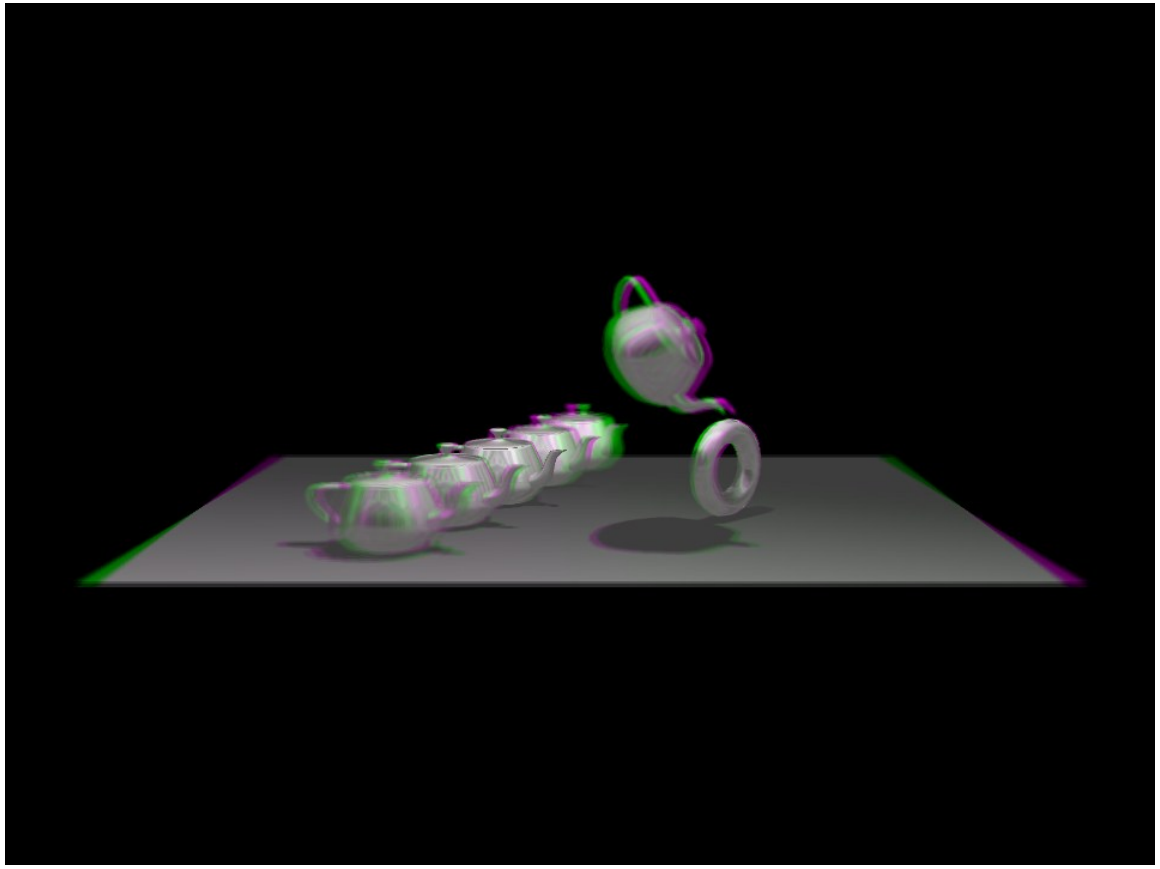
Frustum applications

- Anaglyphs for 3D viewing (i.e., left eye right eye)





Based on slides by Steve Marschner

Review and more information

- FCG chapter 7
  - Viewing and projection

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